
CS246: Database Management Systems Lab

Lab # 13 (1 Questions, 118 Marks)

Lab session: AL1

Held on: 20-Apr-2026 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 4

Submissions accepted from: 16:45 hrs, 20-Apr-2026

Submissions accepted till: 18:00 hrs, 20-Apr-2026

Instructors Dr. V. Vijaya Saradhi

Every one minute delay after 17:30 hrs will result in one mark deduction

Please do not quote Network issues or technical issues after 17:30hrs

Head TA Divanshu Walia

Department of CSE, IIT Guwahati

1. This lab theme is database replication
2. Read the document `database-replication.pdf` and prepare MySQL server for replication.

Question 1: (118 points)

Perform the following tasks.

Task 01a (1 mark) On source server: create database `week13a` and `week13b`

Task 01b (1 mark) On replica server: create database `week13a` and `week13b`. Ensure both databases are empty and identical before replication begins.

Task 02 (10 marks) **Source Server Configuration:** Perform the steps described in the word document. Important: for task 01 to task 10 use the database `week13a`.

Verification: Execute

```
SHOW VARIABLES LIKE 'log_bin';
```

and confirm that binary logging is enabled.

Configure the **source server** using the provided replication guide. Ensure that:

1. server-id is set to 1,
2. Binary logging is enabled, and
3. The database name is correctly specified in `binlog_do_db`

Task 03 (20 marks) **replica Server Configuration:** Perform the steps described in the word document. Ensure correct values of: `SOURCE_HOST`, `SOURCE_USER`, `SOURCE_PASSWORD`, `SOURCE_LOG_FILE`, and `SOURCE_LOG_POS`.

Verification: Execute

```
SHOW REPLICA STATUS\G
```

Ensure:

Replica_IO_Running: Yes

Replica_SQL_Running: Yes

Task 04a (3 marks) On source server create the following tables:

1. (1 marks) The **sailor** table has the following description

column #	name	data type	constraint
1	sid	int	primary key
2	sname	char(50)	None
3	rating	int	None
4	age	decimal(3,1)	None

2. (1 marks) The **boat** table has the following description.

column #	name	data type	constraint
1	bid	int	primary key
2	bname	char(50)	None
3	bcolor	char(50)	None

3. (1 marks) The **reserve** table has the following description.

column #	name	data type	constraint
1	sid	int	refers to sailor
2	bid	int	refers to boat
3	day	date	None
sid, bid, day is primary key			
sid foreign key cascade on update			
sid foreign key cascade on delete			
bid foreign key cascade on update			
bid foreign key cascade on delete			

Task 04b (3 marks) On replica server: Demonstrate through SQL query the replicated tables

Task 05a (3 marks) On source server:

1. Insert data from **sailors.csv** into **sailor** table
2. Insert data from **boats.csv** into **boat** table
3. Insert data from **reserves.csv** into **reserve** table

Task 05b (3 marks) On replica server: Demonstrate through SQL query the replicated data in the respective tables.

Task 06a (6 marks) On source server:

1. (1 mark) Alter table **sailor** to add one column **earning** int with default value 100.
2. (1 mark) Alter table **boat** to add one column **make** char(20) with default value **Toyota**.
3. (1 mark) Alter table **reserve** to add one column **amount** int with default value 10.
4. (1 mark) Insert one row with random data into **sailor** table
5. (1 mark) Insert one row with random data into **boat** table

6. (1 mark) Insert one row with random data into **reserve** table

Task 06b (3 marks) On replica server: Demonstrate through SQL query the replicated data in the respective tables.

Task 07a (3 marks) On source server:

1. (1 mark) Drop **earning** column from **sailor** table
2. (1 mark) Drop **make** column from **boat** table
3. (1 mark) Drop **amount** column from **reserve** table

Task 07b (3 marks) On replica server: Demonstrate through SQL query the replicated data in the respective tables.

Task 08a (4 marks) On source server:

1. Write a stored procedure **insert_sailors** which takes no arguments and inserts 10000 records of sailors into **sailor** table. Use deterministic values (e.g., incremental IDs).

Task 08b (4 marks) On replica server: Demonstrate display the replica status through **SHOW REPLICA STATUS \G**. Observe the fields

Task 09a (4 marks) On source server:

1. (4 marks) Write a trigger on **boat** table and after insert, this trigger should insert one record in **reserve** table for the newly inserted boat and **sid** must be selected from existing sailors record and a random **day**

Task 09b (2 marks) On replica server: Demonstrate the effect trigger on **reserve** table. Insert one record into the boat table on the source server. Verify the following:

1. The trigger is executed on the source:
Check that one corresponding record is inserted into the reserve table.
2. Replication correctness:
Verify that the inserted record in boat and the corresponding record in reserve are both reflected on the replica server.
3. Consistency check:
Ensure that the values of bid, sid, and day in the replica match exactly with those in the source.

Task 10a (3 marks) On source server: Create a view **sailor_reserve_boat** containing columns **sname**, **bname**, **day**

Task 10b (1 mark) On replica server: Show the view.

Task 11a (10 marks) **Replication Filtering:** This task involves sharing only one table of **week13b** database. **Source Server Configuration:** Perform this task only after completing task 01 to task 10. Do not attempt task 11 as your first task. For task 11 use the database **week13b**.

1. (3 marks) Create three tables **sailor**, **boat**, **reserve** identical to Task 03a description and populate data from files **sailors.csv**, **boats.csv**, **reservers.csv**
2. (3 marks) Create a table **reserve_details** having the following description

column #	name	data type	constraint
1	sname	char(50)	sname, bname, day is primary key
2	bname	char(50)	
3	day	date	

3. (7 marks) Edit the file `/etc/mysql/mysql.conf.d/mysqld.cnf` using root privileges as follows

```
server-id = 1
log_bin = /var/log/mysql/mysql-bin.log

# Replication filters
binlog-do-db = week13b
binlog-do-table = week13b.reserve_details
binlog-ignore-db = mysql
binlog-ignore-db = information_schema
binlog-ignore-db = performance_schema
```

4. (3 marks) Restart MySQL.
5. (1 mark) Randomly update one record of **sailor** table
6. (1 mark) Randomly delete one record of **sailor** table
7. (1 mark) Randomly update one record of **boat** table
8. (1 mark) Randomly delete one record of **boat** table

Task 11b (5 marks) On replica server:

1. Show the contents of **reserve_details**
2. Show the contents of **reserve_details** after updating **sailor** table on source server
3. Show the contents of **reserve_details** after deletion of a record in **sailor** table on source server
4. Show the contents of **reserve_details** after updating **boat** table on source server
5. Show the contents of **reserve_details** after deletion of a record in **boat** table on source server

Task 12 (3 marks) Stop replication on replica.

Task 13 (3 marks) Stop MySQL sever on Source and replica.